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Assignment - 2

CSA0612

DESIGN AND ANALYSIS OF ALGORITHM
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69.

Sorting Techniques - Detailed Comparison of selection sort and Bubble sort.

problem interpretation:

→ sort the dataset [48, 22, 35, 10, 27, 16]
in ascending order using:

* selection sort

* Bubble sort.

show each pass, intermediate arrays.

Comparisons, swaps, and compare both algorithms

selection sort:

→ selection sort finds the smallest element in the unsorted portion and places it at the beginning.

Initial Array:

[48, 22, 35, 10, 27, 16]

Pass	Array After Pass
Initial	48, 22, 35, 10, 27, 16
Pass 1	10, 22, 35, 48, 27, 16
Pass 2	10, 16, 35, 48, 27, 22
Pass 3	10, 16, 22, 48, 27, 35
Pass 4	10, 16, 22, 27, 48, 35
Pass 5	10, 16, 22, 27, 35, 48

Final sorted Array.

[10, 16, 22, 27, 35, 48]

For 6 elements:

$$= 5 + 4 + 3 + 2 + 1 = 15 \text{ Comparisons}$$

Swaps:

pass 1 $\rightarrow$ 1

pass 2 $\rightarrow$ 1

pass 3 $\rightarrow$ 1

pass 4 $\rightarrow$ 1

pass 5 $\rightarrow$ 1

Total Swaps = 5.

Bubble Sort:

Bubble sort compares adjacent elements, and swaps them if they are in the wrong order. The largest element moves to end after each pass.

[10, 16, 22, 27, 35, 48]

Comparisons. $5 + 4 + 3 + 2 + 1 = 15$

1 $\rightarrow$ 5, 2 $\rightarrow$ 3, 3 $\rightarrow$ 1, 4 $\rightarrow$ 1, 5 $\rightarrow$ 0

Total Swaps = 10.

Time Complexity:

Algorithm:

* Selection Sort

* Bubble Sort.

Best Case:

* $O(n^2)$

* $O(n)$ (Optimized).

Average case:

* $O(n^2)$

* $O(n^2)$

Worst Case:

* $O(n^2)$

* $O(n^2)$

Comparison:

Feature:

→ comparisons, swaps, Time

Complexity, memory, Better for.

select sort:

→ 15, 5, $O(n^2)$, $O(1)$, Fewer swaps.

Bubble sort:

→ 15, 10, $O(n^2)$, $O(1)$

Nearly sorted data (Optimized Version).

Justification:

- sorted Array [10, 16, 22, 27, 35, 48]
- Both algorithms perform 15 comparisons
- Selection sort performs only 5 swaps while Bubble sort performs 10 swaps.
- Since selection sort requires fewer swaps, selection sort is preferable for this dataset.

Application:

selection sort:

- * Small datasets
- * Memory constrained systems.
- * When minimizing swaps is important.

Bubble Sort:

- * Educational purpose
- * Nearly sorted data
- * Small lists.